# Our Data Pipeline

# Ex: Synthetic Data Easy QFT

## Example

Consider a complex scalar field $\Phi$ in four spacetime dimensions described by the Lagrangian:

$$\mathcal{L} = (\partial_\mu \Phi^\dagger)(\partial^\mu \Phi) - V(\Phi) \tag{15}$$

with the potential

$$V(\Phi) = \lambda \left( |\Phi|^2 - \frac{v^2}{2} \right)^2 \tag{16}$$

where $\lambda > 0$ and $v > 0$ are real parameters. This theory has a global $U(1)$ symmetry, $\Phi \to e^{i\alpha}\Phi$.
The potential induces spontaneous symmetry breaking (SSB). We choose the vacuum expectation value (VEV) to be real: $\langle \Phi \rangle = \frac{v}{\sqrt{2}}$. The quantum fluctuations around this VEV are parameterized by real scalar fields $h(x)$ (the massive mode) and $\pi(x)$ (the massless Goldstone boson) as:

$$\Phi(x) = \frac{1}{\sqrt{2}}(v + h(x) + i\pi(x)) \tag{17}$$

The conserved Noether current associated with the $U(1)$ symmetry is given by:

$$J^\mu(x) = i \left( (\partial^\mu \Phi^\dagger)\Phi - \Phi^\dagger(\partial^\mu \Phi) \right) \tag{18}$$

According to Goldstone's theorem, the massless mode $\pi(x)$ is created from the vacuum by the action of this current. The matrix element of the current between the vacuum $|0\rangle$ and a one-Goldstone boson state $|\pi(p)\rangle$ with momentum $p$ is parameterized by a constant $F$, known as the decay constant:

$$\langle 0 | J^\mu(x) | \pi(p) \rangle = -iFp^\mu e^{-ip\cdot x} \tag{19}$$

**Task:** By expanding the current $J^\mu(x)$ in terms of the fields $h(x)$ and $\pi(x)$, determine the decay constant $F$. Find the numerical coefficient $C$ in the relation $F = C \cdot v$.
**Conventions:** The one-particle states $|\pi(p)\rangle$ are normalized such that the matrix element of the field operator is $\langle 0 | \pi(x) | \pi(p) \rangle = e^{-ip\cdot x}$.

**Goal**

Simple tasks for small models

**Features**

- Ample background context
- Simple Tasks
- One task per problem
- Few Calculations